

DTSA 5013

Generalized Linear Models and Nonparametric Regression

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	Machine Learning Engineer (32.0 / 100)	23.4 / 100
SPECIALIZATION Statistical Modeling for Data Science Applications		

23

CURRENT AVG SCORE
across 6 job roles

58

POTENTIAL SCORE
with recommended additions

+37

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Generalized Linear Models and Nonparametric Regression teaches generalized linear models (GLMs), which provide an introduction to classification (through logistic regression); nonparametric modeling, including kernel estimators, smoothing splines; and semi-parametric generalized additive models (GAMs). Emphasis will be placed on a firm conceptual understanding of these tools. Attention will also be given to ethical issues raised by using complicated statistical models.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE		PROJECTED SCORE		POINT GAIN	PRIORITY
Machine Learning Engineer		32.0		69.2	+37.2	HIGH
Data Scientist		24.9		59.1	+34.3	HIGH
AI Engineer		23.6		59.2	+35.6	HIGH
ML Engineer		21.8		57.4	+35.6	HIGH
MLOps Engineer		21.2		54.9	+33.7	HIGH
Applied Scientist		16.9		48.6	+31.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

✓ Course vocabulary is well-aligned with industry job descriptions.

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

✓ This course is already well-aligned with industry expectations. Continue monitoring for emerging tools and frameworks.

GAP KEYWORDS BY ROLE

Machine Learning Engineer	32.0 → 69.2 (+37.2 pts)
data predictive learning design development science model applications	
Data Scientist	24.9 → 59.1 (+34.3 pts)
data presentation different behaviors build problems	
AI Engineer	23.6 → 59.2 (+35.6 pts)
data learning build model problems machine design make	
ML Engineer	21.8 → 57.4 (+35.6 pts)
data learning machine model build problems global digital	
MLOps Engineer	21.2 → 54.9 (+33.7 pts)
data model learning machine development error	
Applied Scientist	16.9 → 48.6 (+31.7 pts)
computational diffusion problems data learning design development protein	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.